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Algorithmic Training

Concept Questions & Answers- Class

Exercises:

1. The investment cycle consists of asset allocation, portfolio construction, implementation, and portfolio attribution. What is the primary role of the trader as part of the investment cycle?
 - The primary role of the trader as part of the investment cycle is implementation.
 - The trader will do the trading in the market.
 - The trader is required to select the appropriate algorithm and parameters (e.g., POV rate) to execute and implement the trade.
 - The trader needs to make sure that the trading decision is consistent with the investment objective of the fund.
2. In the financial industry, trading algorithms are often given fun and entertaining names. But unfortunately, these names do not often adequately describe what the algorithm is trying to accomplish or how it will trade. Describe the algorithmic classification scheme from this chapter and how it helps traders determine what the algorithms are trying to accomplish.
 - Aggressive – an aggressive algorithm will transact at a faster rate by utilizing more market orders than limit orders and utilizing all sources of liquidity including exchanges, venues, and dark pools.
 - Working Order – a working order algorithm will balance the trade-off between market impact and timing risk. This algorithm will determine the appropriate mix of limit and market orders based on their level of risk aversion and order characteristics.
 - Passive – a passive algorithm will transact at a slower rate and will tend to trade using more limit orders and by routing orders to dark pools.
3. Describe each of the nine (9) transaction cost components from the chapter and provide an example of how they may arise during trading.
 - 1. Commission = the broker's commission. This is to compensate the broker for their service during the transaction and usage of their algorithms.
 - 2. Fees & Rebates = the charges levied by trading venue when orders are matched. To attract liquidity into their system, some venues may offer rebates to some participations (e.g., participants entering orders and providing liquidity via a limit order) and charge fees to other participants (e.g., participants taking liquidity via a market order).
 - 3. Bid-Ask Spread = the difference between ask price and bid price. The Bid-Ask spread represents the round-trip cost of a trade (e.g., buying at the Ask Price and selling at the Bid Price will cause the investor to incur a cost equal to the Bid-Ask spread).
 - 4. Taxes = SEC taxes levied on sell orders. May also include income tax and capital gains tax.

- 5. Delay Cost = price movement in the stock between the time of the investment decision and the time the order was entered into the market. While this cost can be positive or negative, it is often a positive cost due to numerous investors uncovering the same opportunity at around the same time.
 - 6. Price Appreciation = the natural price movement of stock. This is also often referred to as momentum and/or alpha.
 - 7. Market Impact Cost = the price movement in the stock caused by the trade or the order. Market impact cost is comprised by two components: a temporary component due to the liquidity needs of the investor and a permanent component due to the information content of the trade.
 - 8. Timing Risk = the uncertainty surrounding the expected trading cost estimate. Timing risk is comprised primarily of price volatility but also include a volume uncertainty component and estimation error of the market impact model.
 - 9. Opportunity Cost = the missed profiting opportunity that arises when the portfolio manager is not able to complete the entire order.
4. Describe a VWAP algorithm and a TWAP algorithm. How are they similar? How are they different?
- VWAP & TWAP are trading algorithms that employ time slicing strategies.
 - VWAP will slice the order based on historical trading patterns of the stock. Different share quantities will be executed in each time interval.
 - TWAP will slice the order based on time intervals where the same number of shares will be executed in each time interval.
5. What is a POV algorithm? What are the advantages of a POV algorithm compared to a VWAP algorithm? What are the disadvantages of a POV algorithm compared to a VWAP algorithm?
- A POV algorithm is an algorithm that participates with actual market volumes.
 - A VWAP algorithm slices an order based on historical intraday trading patterns.
 - A POV algorithm allows investors to take advantage of increased market liquidity by trading more shares during the time period and will not force trades in times of insufficient liquidity.
 - A POV algorithm may not complete the order if there is insufficient liquidity on the day and the trader does not increase the POV rate. For example, if an investor has an order that is 10% of ADV and wishes to trade the order in the market utilizing a POV rate of POV=10% the order will be completed before the end of the day (on a normal volume day). However, if there is less volume on the day, e.g., only one-half of the normal volume, then the order will not be completed as the original specified POV rate.
6. An index investor with a large order is trying to execute shares at the closing price on the day. From the list of algorithms in the section titled “Types of Trading Algorithms,” which algorithm would you select and why?
- The index investor would select a back-loaded algorithm and trade into the close.
 - This will minimize the price impact of the closing price.

- This algorithm would start trading slow and gradually increase its trading rate into the close.
7. An active manager has determined that stock RLK is appropriately priced in the market but has a very strong long-term growth potential. The manager decided to purchase a large block of RLK shares. From the list of algorithms in the section titled “Types of Trading Algorithms,” which algorithm would you select and why?
- Since stock RLK is appropriately priced in the market the active manager does not have an opportunity to uncover any value. E.g., the stock is not misvalued.
 - Optimal = the active manager may elect to trade the order using a working order algorithm such as an Optimal Algorithm that balances the trade-off between market impact and timing risk.
 - Passive = since the stock is appropriately valued and has strong long term growth prospects, the active manager may elect to trade this order at a slower rate (e.g., passive) to minimize market impact cost.
8. Describe a Displayed “Lite” Venue. Describe a Dark Pool. Why would an investor choose to trade in a displayed market such as an exchange or ATS? Why would an investor choose to trade in a dark pool?
- Lite Venue (also called a Displayed Market) = orders are displayed and disseminated to the public.
 - Use a lite venue when you want to trade via a limit order and have this order disseminated publicly to attract counter parties to execute the trade. Your order would be entered into the order book of the venue.
 - Use a lite venue when you wish to trade at a displayed quote such as executing via a market order.
 - Dark Pool = order and price information is not publicly displayed or disseminated to the public.
 - Use a dark pool when transacting a large order and you do not wish to have the details of this order made known to the public.
9. Describe each of the three venue pricing models: commission-based, maker-taker, inverted maker-take. Why would an investor choose to use a maker-taker venue? Why would an investor choose to use an inverted maker-taker model? Why would an investor choose to use a commission-based model?
- Commission-Based = both buyer and seller pay the venue a commission when a trade occurs.
 - Maker-Taker = The party posting liquidity (e.g., entering a limit order into the venue) receives a rebate and the party taking liquidity (e.g., using a market order) is charged a fee. An investor would use this venue when they wish to trade via a limit order, and they expected their order to be transacted
 - Inverted Maker-Taker = The party posting liquidity (e.g., entering a limit order into the venue) pays a fee and the party taking liquidity (e.g., using a market order) receives a

rebate. An investor would use this venue if there are a large number of orders already entered into the venue at their desired price and they are concerned that their order will not be executed. This venue in a way moves their order to the top of the queue.

10. Describe each of the following algorithm order slicing strategies i) time-based, ii) volume-based, and iii) price-based.

- Time-Based = slice the order based on time intervals such as 30-minute or 15-minute intervals. The slicing strategy can be based on historical intraday volume profiles (e.g., VWAP algorithm) or equal weighted (e.g., TWAP algorithm). The time slicing strategy may also be determined via an optimization process (such as an Optimal algorithm).
- Volume-Based = order is sliced based on an order participation rate such as a POV algorithm. For example, a POV=10% algorithm will need to participate with 10% of the market volume. For every 1000 shares traded in the market the POV=10% algorithm will need to transact 100 shares.
- Price-Based = the algorithm will slice the order and trade either faster or slower based on actual market prices and side of the order. An investor with a buy order may decide to trade faster when prices decrease and trade slower when prices increase (if they expect price reversion to occur).

Questions:

11. In our training session we discussed two types of implementation tactics: agency execution and capital commitment (principal bid transaction).

- Briefly describe each.
 - Agency = broker executes the order using best efforts. Execution price is not known in advance. The investor incurs all market risk. The investor pays a lower commission for this service.
 - Capital Commitment = broker provides the investor with a guaranteed execution price. The investor transfers all market risk to the broker. The broker charges a higher fee for this service. The broker will only earn a profit if they are able to offset the position in the market at a cost less than the fee charged to the investor.
- If you are a trader who needs to transact at a known market price and is not allowed to incur any market risk during trading, which implementation tactic would you select and why?
 - Capital Commitment (see definition above)
- If you are a trader who is looking for a low-cost commission fee trade and is comfortable trading in the market at market prices over the trading horizon, which implementation tactic would you accept and why?
 - Agency Transaction (see definition above)

12. In our session we discussed three types of trading algorithm categories: 1) execution algorithms, 2) profit-seeking algorithms, and 3) high frequency trading algorithms.

- i. Which type of trading algorithm would you select if you are asked to execute a large block trade that was specified by your portfolio manager? Why?
 - Execution Algorithm = the order was specified outside of the algorithm by the portfolio manager. Therefore, the only task left is to transact (execute) this decision in the market.
- ii. Which type of algorithm would you select if your company has decided to start an electronic market making desk?
 - High Frequency Trading / Liquidity Algorithm = a liquidity algorithm will seek to profit by liquidity trading, e.g., buying at the bid price (low) and selling at the ask price (high). This is the same strategy that market makers would use prior to algorithmic trading. But now this function is performed by computer algorithms instead of human traders transacting manual orders.
- iii. If you are a quantitative manager employing a statistical arbitrage and pairs trading algorithm, which type of algorithm would you select?
 - Profit Seeking Algorithm = this type of algorithms will determine when it is appropriate to buy or sell and at which price, and will then transact that decision. A profit seeking algorithm makes the investment decision and performs the actual transactions (trading).

13. We introduced an algorithmic decision-making process to assist investors select the execution algorithm and specify algorithmic parameters?

- i. What are the key decisions that need to be made by the portfolio manager? How does the investment decision of the fund manager effect these decisions?
 - Macro = how the order is to be traded by the algorithm. This specification needs to be consistent with the investment objective of the fund.
 - Micro = how the algorithms is to adapt to changing market conditions (volumes, price, and volatility) during the day.
- ii. What decisions are made by the algorithm? How does real-time market conditions effect how the algorithm will trade?
 - LOM = uses a limit order model to determine the appropriate mix of limit and market orders.
 - SOR = determines which venues to route the order for execution.

14. You are the portfolio manager of a large fund and concerned about recent high transactions costs due to delay cost and opportunity cost.

- i. How can a portfolio manager minimize the delay cost component?
 - By knowing which brokers and algorithms provide the best execution quality for which types of orders. This would allow the buy side trader to route the order to the

appropriate broker and specify the execution algorithm to use for the order without any delay.

- ii. How can a portfolio manager minimize the opportunity cost of a trade?
 - By knowing how many shares of the stock can be absorbed into the market at the specified price and cost constraints.

15. You are an index manager and need to purchase a large block order for a stock that is being added to your benchmark index at the end of the day.

- i. How would you purchase these shares if you do not have any expectations short-term movement of the stock? What would you select as your benchmark price?
 - Trade using a back-loaded strategy into the close combined with a MOC algorithms. Because we need to purchase a large block order, the price impact of trading all shares on the close with a MOC algorithm would likely be very high. Therefore, we would use a back-load strategy where we start off trading slower and gradually increase our trading rate into the close, as well as trading in the closing auction.
 - Closing Price as the benchmark price because Index funds are valued using the closing price on the day.
- ii. The company that is going to be added to your benchmark index at the end of the day is going to hold a press conference in the afternoon to provide the market with an update on company performance. You believe that the company is going to announce a new product that they have been working on and expect the market to react very favorably and you expect to see an increase in price. How would this information change how you will purchase shares? What is the risk associated with this decision?
 - The portfolio manager believes that there is going to be favorable news about the company accounted in the afternoon which would cause the stock price to increase.
 - Since the stock is going to be added to the index at the end of the day the stock price will reflect the price increase.
 - In this case, the portfolio manager would prefer to buy the shares before the company new announcement which will result in an increase in price.
 - If the portfolio manager is correct, this decision will result in an increase in portfolio value due to the company new and increase in stock price.
 - If the portfolio manager is incorrect, and the new is not favorable, the portfolio just increased their tracking error risk. If the new is not favorable, and the stock price decreases, the portfolio manager would have bought shares at a higher price during the day. Thus, resulting in a decrease in the portfolio values.
 - Purchasing shares before the company announcement will result in increased tracking error risk.

16. You are an Active Manager who has uncovered a stock that is currently undervalued in the market and will likely increase to its fair value by the end of the day. Based on this research, you decide to purchase shares while the stock price is undervalued and then sell the shares after the market realizes the mispricing and the stock price increases to its fair-value price at the close.

- i. How would you execute these shares? Specify an algorithm that you would select to execute this order.
 - Aggressive algorithm that would trade at a fast rate to purchase as many shares as possible before the stock increases to its fair value price. You can then sell the shares you purchased at the end of the day at the higher price.
 - ii. During the trading day, the price of this stock further decreases but your expectations remain the same. How would this change your investment and trading decision.
 - Increase the trading rate further so that you can transact at the lower (and better) market prices. You can then sell the shares you purchased at the end of the day at the higher price.
17. You are a Quantitative Portfolio Manager who has just rebalanced the portfolio using optimization techniques. The basket consists of buy and sell orders.
 - i. What type of algorithm would you use to execute this basket of stock?
 - A working order algorithm that balances the trade-off between market impact cost and timing risk. Because the basket consists of buy and sell orders it is somewhat protected from market movement.
 - ii. How would you specify your level of risk aversion?
 - The level of risk aversion should be set at the same level of risk aversion that was used in the portfolio optimizer. This value is also the Sharpe Ratio of the fund.
18. You are the portfolio manager of a large institution. Describe how your trading desk would utilize each of the following broker dealer trading desks:
 - i. Cash Desk
 - Use the cash desk when trading single stock orders (that are not part of a basket) and where the primary concern is adverse price movement.
 - ii. Program Desk
 - Use the program desk when trading groups of stocks known as baskets, portfolios, programs, or trade lists, and where the primary concern of trading is risk management of the basket.
 - iii. Electronic Desk
 - Use the electronic desk when you wish to remain anonymous and wish to have full control of your order so that you can select and revise algorithms and algorithmic parameter during the trading day.
19. You are a portfolio manager for a large fund. Your research team has uncovered a stock that has very strong long-term growth potential and is also under-valued in the market. Historically, this stock has exhibited short-term price momentum patterns. For example, if the price increases it is likely to continue to increase over the entire day. Based on this information, you decide to

purchase a large block of these shares. As a portfolio manager, you want to make the best implementation decisions possible so to maximize portfolio performance. Therefore, you follow the algorithmic decision-making process from this chapter.

- i. As a portfolio manager, how would you specify your macro and micro decisions to execute this order?
 - Select the macro strategy to be aggressive because the stock is currently undervalued, and the fund would wish to purchase shares because the rest of the market learns that the stock is undervalued.
 - Select the micro strategy to adapt to changing market conditions and the price momentum patterns. Increase the trading rate to transact even quicker if the price starts to increase because it is expected that once the stock price starts to increase it will continue to increase through the remainder of the day.
 - ii. If the stock has historically exhibited mean-reverting patterns when an increase in stock price was followed by decrease in price shortly thereafter, how would this information change how you specify your macro and micro decisions?
 - Macro strategy = keep the same strategy as above.
 - Micro strategy = adapt to changing prices by trading slower as the price increases because the stock exhibits price reversion. Therefore, slow down the trading until the price reverts to the lower price. If the price decrease, in this case, begin to trade faster to lock in the lower prices before the price trend reverses and increases back to the higher price. Therefore, if prices increase, slow down and wait for prices to come back down. If prices decrease from you increase your trading rate to lock in the lower prices.
20. In this chapter we decreased Transaction Cost Analysis and analytics. Describe how each of the following can be used during the implementation phase of the investment cycle?
- i. Pre-Trade Analysis
 - Used to estimate trading costs and evaluate alternative trading strategies and algorithms to determine which was most appropriate for the order.
 - ii. Intra-Day Analysis
 - Used to continuously measure trade prices and cost of your order and to help determine when it is appropriate to revise the algorithms and take advantage of changing market conditions.
 - iii. Post-Trade Analysis
 - Use to measure trading costs and evaluate broker and algorithmic performance. This helps determine which brokers and algorithms are best for which types of orders and market conditions.